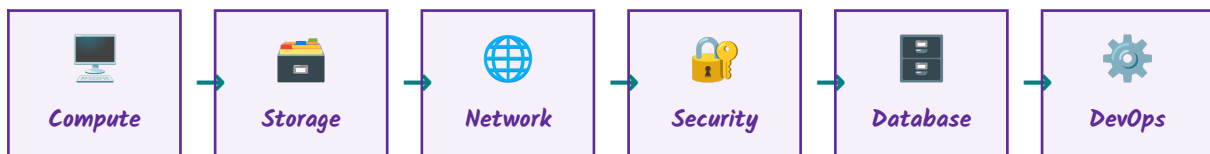




AWS

Complete Cloud Computing Notes

From Virtualization to Production — Digitized & Organized



These notes connect AWS's core services into one complete picture — compute, storage, networking, security, databases, and DevOps, all in one place. ★

Topics Covered

Compute (EC2, Lambda, ECS, EKS)	Storage (S3, EBS, EFS, FSx)	Networking (VPC, ELB, Route 53)	Security & IAM
Databases (RDS, DynamoDB, Redshift)	DevOps (CI/CD, Terraform, IaC)	Pricing & Cost Tools	Messaging (SQS, SNS, EventBridge)
Monitoring (CloudWatch, CloudTrail)	AI/ML Services	Governance & Compliance	Migration & Hybrid Cloud

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1. Virtualization

1.1 What is a Virtual Machine?

A computer is fundamentally hardware. Virtualization is the process of setting up that hardware so it can run multiple operating systems on top of it — for example Linux OS and Windows OS running side by side on the same physical machine.

Hypervisor

A software that creates and runs virtual machines. It sits between the physical hardware and the virtual machines.

- VirtualBox-style hypervisors share hardware resources from the Host OS.
- Each VM gets a separate set of virtual CPU, RAM, storage, etc.
- VMs are fully isolated — independent of the host OS.

1.2 Benefits of Virtualization

- You don't need new physical resources to run a different OS.
- No risk of issues affecting your primary OS.
- Easy to test applications across different operating systems.

1.3 Types of Hypervisor

Type 1 — Bare Metal Runs directly on the hardware. Examples: AWS, VMware ESXi	Type 2 — Hosted Runs on top of a Host OS. Example: VirtualBox on Windows/macOS
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1.4 Why Companies Use Virtualization vs. Physical Cores

- Cheaper than buying separate physical machines.
- Reduces workload, physical space, and energy consumption.
- Easy backup using snapshots.

- Easy recovery.

2. Cloud Computing Fundamentals

Cloud Computing

On-demand delivery of IT resources over the Internet with pay-as-you-go pricing. You access computing resources (like servers, storage, databases, and software) over the internet, rather than owning and maintaining physical hardware yourself.

2.1 Cloud Service Providers

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud Platform (GCP)
- IBM Cloud
- Oracle Cloud
- Salesforce
- VMware Cloud

2.2 Types of Cloud Computing (Service Models)

- IaaS — Infrastructure as a Service
- PaaS — Platform as a Service
- SaaS — Software as a Service

Infrastructure as a Service (IaaS)

IaaS contains the basic building blocks for cloud IT. It typically provides access to networking features, computers (virtual or on dedicated hardware), and data storage space. IaaS gives you the highest level of flexibility and management control over your IT resources. It is most similar to the existing IT resources with which many IT departments and developers are already familiar.

Real-life analogy

Only the shell of a building is made — you fit it out yourself. You get the raw plot/structure and decide everything else.

Platform as a Service (PaaS)

PaaS removes the need for you to manage the underlying infrastructure (usually hardware and operating systems) and allows you to focus on the development and management of your applications. This helps you be more efficient, as you don't need to worry about resource procurement, capacity planning, software maintenance, patching, or any of the other undifferentiated heavy lifting involved in running your application.



Real-life analogy

You just need a product — the building's floors/shops are already made; you simply move your product in.

Software as a Service (SaaS)

SaaS provides you with a complete product that is run and managed by the service provider. In most cases, people referring to SaaS are referring to end-user applications (such as web-based email). With a SaaS offering, you don't have to think about how the service is maintained or how the underlying infrastructure is managed; you only need to think about how you will use that particular software.



Real-life analogy

A vending machine — fully ready to use, nothing to set up.

2.3 Cloud Deployment Models

- Public Cloud
- Private Cloud
- Hybrid Cloud



Public Cloud

A shared cloud environment where multiple users can access services over the internet, like AWS or Azure.

Private Cloud

A dedicated cloud environment for one organization, offering more control and privacy.

Hybrid Cloud

A mix of public and private clouds, allowing data and applications to move between them for flexibility.

2.4 Common Features Offered by Cloud Providers

- Compute services
- Storage services
- Database services
- Networking services
- Container services
- Machine learning services

3. AWS Overview

AWS (Amazon Web Services)

AWS is a cloud computing / cloud service provider. It offers a pay-as-you-go model, allowing you to pay only for what you use, which is ideal for optimal cost.

- AWS was launched in 2006 as the first market-leading cloud platform.
- Started with S3 for storage and EC2 for compute power.
- Now expanded into offering over 200 fully-featured services across various domains like AI, Machine Learning, and IoT.

- Used by: Netflix, NASA.

3.1 What Makes AWS So Special

- Scalability
- Global Reach
- Security
- Reliability

4. AWS IAM (Identity & Access Management)

IAM is a service that helps you securely control access to AWS resources. It allows you to manage users, roles, and permissions to define who can access what within your AWS environment.

- IAM is a free service — no additional cost.
- It's a global service.
- A root account is created by default — it should not be used or shared.
- Create users.
- Create groups.

4.1 AWS IAM Best Practices

- Avoid using the root account except for initial account setup.
- Use a password policy or MFA (Multi-Factor Authentication).
- Use access keys for CLI access.
- Add users to a group and assign permissions to the group.
- Never share access keys or passwords.



Exam tip

Root account = the master login for the whole AWS account. Lock it down with MFA and never use it for daily tasks — create an IAM user with the right permissions instead.

5. Amazon EC2 (Elastic Cloud Compute)

Key point

Configuration has to be done yourself (खुद configuration करनी होती है) — EC2 is not fully managed like some other services.

EC2 is a cloud service that provides resizable virtual servers, called instances, which you can use to run applications.

Example

Imagine you're running a business and need a server to host your website or application. Instead of buying and managing a physical server, AWS EC2 lets you rent virtual servers in the cloud. These virtual servers are called instances.

5.1 What You Can Configure on Launch

- OS
- RAM
- Disk space
- Network
- How to access the machine

Instance Type

Select the hardware capacity (e.g., CPU, memory) for your instance.

AMI (Amazon Machine Image)

Choose the operating system and software for your instance (e.g., Linux, Windows).

Storage

Configure the type and size of storage (e.g., EBS volumes).

 Security Groups

Set up firewall rules to control inbound / outbound traffic.

 Key Pair

Create or use an existing key pair for SSH access.

 Network Settings

Configure VPC, subnet, and assign public or private IP addresses.

 IAM Role

Attach an IAM role for permission to access other AWS resources.

 User Data

Add a script to be executed when the instance starts.

 Elastic IP

Optionally associate a static IP address for consistent public access.

5.2 EC2 Security Groups — Important Points

- Region-specific.
- Only "Allow" rules exist (there is no "Deny" rule).
- All inbound traffic is blocked and outbound traffic is allowed by default.

You define rules as specific:

- IP addresses or ranges (e.g., allow traffic only from a specific IP or range of IPs).



Good to remember

If you allow incoming traffic on a specific port (e.g., port 80 for HTTP), the outgoing response traffic is automatically allowed without an explicit outbound rule — Security Groups are stateful.

6. Common Ports You Should Be Aware Of

Protocol	Port	Purpose
HTTP	80	Unencrypted web traffic
HTTPS	443	Encrypted web traffic (SSL / TLS)
SSH	22	Secure remote access to a server (via terminal)
FTP	21	File Transfer Protocol (unsecured)
SFTP	22	Secure File Transfer Protocol
SMTP	25	Simple Mail Transfer Protocol (email sending)
RDP	3389	Remote Desktop Protocol (Windows remote access)
MySQL	3306	MySQL database connections
PostgreSQL	5432	PostgreSQL database connections
DNS	53	Domain Name System (converts domain to IP address)

6.1 How to SSH into an EC2 Instance

SSH allows you to control / access a remote machine.

7. AWS EBS (Elastic Block Store)

EBS (Elastic Block Store)

A cloud-based storage service that provides durable, high-performance block storage for use with Amazon EC2 instances.

It works like a virtual hard drive, allowing you to store and access data even when your EC2 instance is stopped or terminated.

Example

If you're hosting a MySQL or PostgreSQL database, you need reliable, high-performance storage to handle frequent read/write operations. EBS provides persistent, fast storage that ensures your data is saved even if the EC2 instance is restarted — making it ideal for databases.

7.1 Important Points About EBS Availability Zone

- Region and AZ (Availability Zone) specific.
- Built-in redundancy — EBS volumes are automatically replicated within the same Availability Zone to prevent data loss due to hardware failure.
- Different volume types: gp2/gp3, io1/io2, st1, sc1.
- Allows encryption and snapshots for backup.
- Scalable — volumes can be resized.
- No data loss while resizing.

7.2 EBS Snapshots

What if we want to copy our data to:

- A new AZ (Availability Zone)
- A new Region

**Exam tip**

An EBS Snapshot is a point-in-time backup of a volume, stored in S3 behind the scenes. Snapshots are the way to move EBS data across Availability Zones or Regions — you snapshot, then create a new volume from that snapshot wherever you need it.

8. AWS AMI (Amazon Machine Image)

**AMI (Amazon Machine Image)**

A pre-configured template that provides the necessary information to launch an EC2 instance in AWS.

It includes:

- Operating System
- Application server
- Pre-installed and configured software

8.1 Types of AWS AMI

**Public AMIs**

Available to all AWS users. Useful for basic use cases like popular operating systems (e.g., Ubuntu, CentOS).

**Private AMIs**

Created by a user and only available within that account or shared with specific accounts.

**Paid AMIs**

Provided by third parties through the AWS Marketplace, offering software like databases or web servers.

8.2 EC2 Image Builder

- Automates VM or image creation — creation, testing, and deployment of AMIs.
- Can be configured to run at regular intervals (e.g., daily, weekly, or monthly).
- Free to use.

9. AWS ELB (Elastic Load Balancing)

ELB (Elastic Load Balancing)

A service that automatically distributes incoming application traffic across multiple targets, such as EC2 instances.

- ELB = Elastic Load Balancing
- ASG = Auto Scaling Group

9.1 Scalability

Scalability means the ability to grow your system's resources when your application or website gets more traffic or more users.

10. Scalability, Availability & Elasticity

10.1 Vertical Scalability (Scaling Up)

Vertical scalability means adding more power (CPU, RAM) to your existing server.



Example

t2.micro → t2.large

10.2 Horizontal Scalability (Scaling Out)

Horizontal scalability means adding more instances to distribute the load.

- You can add more EC2 instances behind a load balancer.

10.3 High Availability (HA)

High Availability (HA) means keeping your service up and running with minimal downtime so it's always accessible to users.

- Example: running resources in multiple Availability Zones (AZs).

10.4 Elasticity

Elasticity means the ability to automatically adjust resources as the demand changes — adding more when needed and removing when it's no longer necessary.

11. AWS ELB — How It Works

11.1 What ELB Does

Distributes Traffic

Splits incoming traffic across multiple servers so no single server gets overloaded.

Improves Availability

If one server goes down, the load balancer automatically sends traffic to the working servers, ensuring your application stays available.

Scales Resources

Helps manage high demand by adding more servers during peak times and distributing the load.

- Single point of access, not to be exposed.
- HA (High Availability) across AZs.

11.2 Types of Load Balancers

AWS offers different types of load balancers depending on your need:

Application Load Balancer (ALB)

Perfect for web applications, handling complex HTTP and HTTPS requests (Layer 7).

Network Load Balancer (NLB)

Designed for high-performance and low latency, perfect for TCP/UDP traffic (e.g., gaming, financial apps) (Layer 4).

Gateway Load Balancer (GWLB)

Helps deploy, scale, and manage third-party virtual appliances, such as firewalls and monitoring solutions.

12. AWS Auto Scaling Group (ASG)

ASG (Auto Scaling Group)

A service that automatically adds or removes EC2 instances based on demand to ensure your application is always available.

It helps scale up when more capacity is needed and scale down during low usage to save costs, keeping the right number of servers running at all times.

13. ASG (continued) — Steps to Create an Auto Scaling Group

- Launch template or configuration
- Create Auto Scaling group
- Select VPC and subnets
- Attach load balancer (optional)
- Configure scaling policies
- Health checks
- Add notifications (optional)
- Review and create

14. AWS S3 (Simple Storage Service)

S3 (Simple Storage Service)

A cloud-based storage service that allows you to store, manage, and retrieve large amounts of data like files, images, videos, and backups securely and at scale.

It provides highly reliable, scalable object storage, making your data accessible from anywhere, anytime via the internet.

14.1 S3 Bucket Basics

- Stores data as objects.
- Globally unique name.
- Region specific.
- Each object within a bucket is stored as a key-value pair.

🔑 Maximum Object Size

STB (Terabytes) is the maximum size for a single object.

14.2 S3 Bucket Policies

JSON-based access control policies that you attach directly to an S3 bucket to manage permissions for accessing the bucket and its objects.

They allow you to define who can access the data and what actions they can perform, such as read, write, or delete, enabling fine-grained control over the security of your data stored in S3.

14.3 S3 Versioning

It allows you to keep multiple versions of an object in the same bucket, providing protection against accidental deletion or overwrites.

When versioning is enabled, S3 stores every version of an object, allowing you to recall older versions if needed — making it ideal for data safety and backups.

14.4 S3 Replication

It allows you to automatically copy objects from one S3 bucket to another, which can be:

- Within the same region
- In a different region

It's commonly used for compliance, redundancy, and to improve data access performance by maintaining copies closer to your users.

14.5 S3 Storage Classes

Storage Class	Best For
S3 Standard	Frequently accessed data
S3 Intelligent-Tiering	Data with unknown or unpredictable access patterns
S3 Standard-IA	Infrequently accessed but needs rapid retrieval

<i>S3 One Zone-IA</i>	<i>Non-critical, infrequently accessed data</i>
<i>S3 Glacier</i>	<i>Archival data, rarely accessed</i>
<i>S3 Glacier Deep Archive</i>	<i>Deep archival data, almost never accessed</i>

- *S3 Outposts — local storage (on-premises S3-compatible storage).*

14.6 S3 Bucket Lifecycle

You can use lifecycle policies to control the movement of objects between different storage classes, or delete them entirely, based on specific conditions like age or inactivity.

14.7 S3 Snow Family

The S3 Snow Family is a group of physical devices offered by AWS to help move large amounts of data to the cloud when using the internet isn't practical.

These devices are used when there's too much data to upload over a regular connection, or when dealing with remote areas without good internet.

The Snow Family includes:

AWS Snowcone

A small, portable device for a few terabytes of data.

AWS Snowball

A larger device for moving petabytes of data, and can also be used for edge computing.

AWS Snowmobile

A massive truck-sized container used for exabyte-scale data transfers, typically used by big companies moving entire data centers.

15. Amazon S3 Storage Gateway

It is a hybrid cloud storage service that connects on-premises environments to cloud storage in Amazon S3. It helps extend your local storage to the cloud by acting as a bridge.

16. AWS RDS (Relational Database Service)

RDS (Relational Database Service)

AWS RDS is a managed database service that simplifies database setup, operation, and scaling.

Purpose: Handling administrative tasks like backups, patching, monitoring, and scaling.

16.1 Creating an RDS Instance

- *Create an RDS MySQL instance.*
- *Use Free Tier.*
- *Username will be 'admin' and you can set a password (you can't use special characters).*
- *Set the table's access to true to access it from a local or remote server.*
- *Create a security group (and allow port 3306 from anywhere).*
- *After creating, you can find the endpoint (hostname) to connect to the DB.*

16.2 Installing Docker on an EC2 Instance

```
sudo yum install -y docker
sudo service docker start
sudo usermod -aG docker ec2-user
sudo docker pull phpmyadmin/phpmyadmin - mysql-8.0
```

17. AWS Lambda

AWS Lambda

A serverless computing service that lets you run code in response to events without managing servers.

You just upload your code and AWS automatically handles the rest — scaling as needed and only charging for the time your code runs.

17.1 Event-Driven Function

Lambda is an event-driven service, meaning it runs your code in response to certain triggers or events.

These events can come from many different AWS services, like:

- S3 (file uploads)
- DynamoDB
- API Gateway
- CloudWatch

18. AWS CloudFormation

AWS CloudFormation

An Infrastructure as Code service that lets you define, provision, and manage AWS resources in a declarative, template-based format.

Create a document and AWS lets you create infrastructure from it.

Consistency

Define infrastructure in code.

 Automation

Reduce manual work.

 Repeatable

Replicate environments easily.

19. Amazon Route 53

 Route 53

AWS Route 53 is a scalable DNS service for domain registration, traffic routing, and health-checking capabilities.

19.1 DNS (Domain Name System)

It is the internet service that translates human-friendly domain names like `www.example.com` into machine-readable IP addresses.

 Good to remember

Default port for DNS (Domain Name System) service is 53.

20. Amazon CloudFront (CDN)

 CloudFront

AWS CloudFront is a Content Delivery Network (CDN) that speeds up the delivery of web content to users by caching it at servers (edge locations) close to them, improving load times and performance globally.

20.1 Virtual Private Cloud (VPC)

A private, isolated network within the AWS cloud where you can launch and manage your resources securely.

- To securely isolate and control network environments.

20.2 AWS Amplify

Amplify is a platform that simplifies:

- Building
- Deploying
- Hosting full-stack web and mobile apps.

21. AWS Amplify (continued)

- Connects to backend services like user login, data storage, and APIs.
- Offers hosting and automatic updates for apps.

22. Amazon ECS (Elastic Container Service)

ECS (Elastic Container Service)

A cloud-based container management service that allows you to run and manage Docker containers on a cluster of virtual servers.

22.1 Why ECS

It automatically handles:

- Cluster management
- Updating

22.2 ECS Terms

Cluster

A group of tasks and services. Hosts all the resources and infrastructure.

Service

Handles scalability and load balancing of containers.

Task

Represents the running containers of your application.

23. Amazon EKS (Elastic Kubernetes Service)

EKS (Elastic Kubernetes Service)

A fully managed Kubernetes cluster infrastructure service on AWS.

23.1 Kubernetes

An open-source platform for automating the deployment, scaling, and management of containerized applications.

- *Automating deployment*
- *Management of containerized apps*
- *Scaling*

When you deploy Kubernetes, you get a cluster. Two important parts are:

- *Master (Control Plane)*
- *Worker nodes*

23.2 Prerequisites

Before you begin, ensure you have the following prerequisites set up to use Amazon EKS:

- Set up AWS CLI and configure credentials.
- Install kubectl.
- Install kubectl (Kubernetes CLI).

24. Terraform

Terraform

Terraform is an open-source Infrastructure as Code (IaC) tool.

24.1 IaC (Infrastructure as Code)

It enables you to define, provision, and manage infrastructure across multiple cloud providers using a declarative configuration language.

24.2 Terraform Config

- It uses .tf extension.
- Format is HCL (HashiCorp Configuration Language) — a declarative language.
- It supports JSON also.

24.3 Sample Commands

```
terraform init
terraform plan
terraform apply / destroy
```

24.4 Benefits

Consistency

Define infrastructure in code.

Automation

Reduce manual work.

Repeatable

Replicate environments easily.

24.5 State Management

The state file (`terraform.tfstate`) maintains a detailed record of the current state of managed resources.

The state file can be stored locally or remotely. With remote storage options, enabling collaboration by sharing the state across teams and environments.

25. SSL / TLS

SSL (Secure Socket Layer)

Cryptographic protocols designed to provide secure communication over a computer network.

TLS (Transport Layer Security)

The successor to SSL.

25.1 How SSL/TLS Secures Data

Encrypts data and ensures its:

- Integrity
- Confidentiality, and
- Authenticity

...between a client and server.

26. SSL/TLS — Important Terms

Confidentiality

Data is only accessed by the client/server (Encryption).

Integrity

Data is not modified in between (Hashing).

Authentication

Verifying the identity of the parties who they are supposed to be (Certificate).

26.1 Encryption

Converting plaintext information into coded form (ciphertext).

- Symmetric Encryption
- Asymmetric Encryption
- Hybrid Encryption

26.2 Hashing

Hashing is the process of converting data into a fixed-size string of characters, a sequence of numbers and letters.

26.3 CA (Certificate Authority)

A Certificate Authority (CA) is a trusted organization that issues digital certificates to verify the identity of websites and enable secure, encrypted communication over the internet.

CA ensures the authenticity and integrity of the SSL certificates they provide.

27. IAM — Deep Dive

IAM

Identity & Access Management. Identifies what is allowed by them.

Core components:

- User
 - Group
 - Role
 - Policy
-
- Group is a way to organize users. Policy defines permissions.
 - A user can get permissions directly or via group.
 - Roles are for delegation and are assumed.
 - With assumed roles, you can talk to other services.

Root User

Made when you sign up for AWS; has full permissions. Should be avoided and enable

MFA.

27.1 Create New User

- No permissions by default.
- You can create around 5000 accounts (1 max).
- Use policies and prefer groups.
- Amazon Resource Name (ARN) is a regular name.
- Username and password, or access key.

27.2 Create Groups — Example Structure

Admin Policy 1	Developer Policy 2	Tester Policy 3
--------------------------	------------------------------	---------------------------

- Add policy to group.
- Add permission to group.
- User gets permission via group.
- Least privilege possible.

27.3 IAM Roles

Roles are assumed by users, applications, and services.

- Example: an EC2 needs to read from S3 (an application needing permission).
- Grants the permission of a role only, temporarily.

27.4 IAM Policies

- Policies are JSON documents.
- Explicitly denied by default.
- Identity-based or resource-based.

AWS supports 6 types of policies:

- Identity-based policy

- *Resource-based policy*
- *Permission boundaries*
- *Organization service control policy*
- *Access control list*
- *Session policy*

27.5 IAM Best Practices

- *Use MFA (Multi-Factor Authentication).*
- *Rotate keys and passwords.*
- *Do not use root account.*
- *Least privilege permissions.*
- *IAM Access Analyzer: a tool to generate least-privilege policy based on access activity.*
- *Regularly review roles, rules, permissions, policies — conditions like IP range or geo-location can be attached.*

28. Basics of Cloud Computing — Recap

Definition

On-demand IT resources delivered via internet.

Cost Model

Pay-as-you-go replaces capital expenditure.

Service Layer

IaaS, PaaS, SaaS — options for flexibility.

28.1 Key Traits

- *Self-service, elasticity, measured usage.*
- *Global Reach: access services from any connected location.*
- *Innovation Speed: rapid experimentation without hardware lead time.*
- *Shared Responsibility: provider secures infra, customer secures data.*
- *Business Impact: drives digital transformation and agility.*
- *Tech Enabler: foundation for AI/ML, IoT, DevOps pipelines.*

29. Why Adopt Cloud Computing

Cost Efficiency

Converts CapEx to predictable OpEx.

Elastic Scaling

Resources adjust automatically to demand.

Time to Market

Deploy applications in minutes, not months.

Global Footprint

Launch workloads near users worldwide.

Built-in Resilience

Multi-AZ architecture improves uptime.

Security Uplift

Advanced encryption and compliance offerings.

Operational Focus

Offload maintenance to the cloud provider.

Data Insights

Native analytics accelerate decision-making.

Collaboration Boost

Supports remote and distributed teams.

30. AWS Global Infrastructure

Region Count

30+ geographic regions across six continents.

Availability

Each region houses 3+ isolated Availability Zones (AZs).

Low Latency Links

Redundant fiber connects AZs securely.

Edge Locations

600+ sites enable CloudFront & Route 53.

Local Zones

Bring compute closer to metro populations.

Data Residency

Region choice meets compliance needs.

Scalable Growth

Continuous expansion roadmap announced yearly.

Fault Isolation

AZ separation limits blast radius of failure.

Performance Goal

99.99% service uptime design target.

31. AWS Core Service Architecture

31.1 Compute Suite

- EC2, Lambda, ECS, EKS

31.2 Storage Options

- S3, EBS, EFS

31.3 Database Family

- RDS, DynamoDB, Aurora

31.4 Networking Tools

- VPC, ELB, Direct Connect, Route 53

31.5 Security Services

- IAM, KMS, Cognito

31.6 Analytics Stack

- Athena, EMR, Glue, QuickSight capabilities

31.7 Management Layer

- CloudWatch, CloudTrail, Systems Manager

31.8 DevOps Toolkit

- CodeCommit, CodeBuild, CodeDeploy

31.9 Unified Ecosystem

Integrated logging, billing, and monitoring across the whole platform.

32. Amazon EC2 Highlights — Recap

 **Virtual Compute**

Launch instances in minutes with custom AMIs.

Instance Families

General, Compute, Memory, GPU, and Storage-optimized types.

Pricing Models

On-Demand, Reserved, Savings Plans, and Spot instances.

Auto Scaling

Adjust instance count to match workload load.

Security Groups

Stateful firewalls control traffic flows.

Nitro System

Provides near-bare-metal performance and security.

OS Flexibility

Run Linux, Windows, macOS.

Integration Points

Works with EBS, S3, and IAM.

User Data

Automates boot-time configuration scripts.

33. AWS Lambda Essentials — Recap

Serverless Compute

Run code without provisioning servers.

Event Triggers

S3, API Gateway, DynamoDB Streams.

Billing Model

Pay per millisecond of execution time.

Instant Scaling

Concurrency adjusts automatically.

Language Support

Node.js, Python, Java, Go, .NET.

Runtime Limit

Functions execute up to 15 minutes.

Config Control

Memory and timeout tunable per function.

Security Model

IAM roles grant least-privilege access.

Use Cases

Microservices, file processing, real-time APIs.

34. Amazon S3 — Recap

Object Storage Service

Store and retrieve any amount of data.

Scalable and Durable

99.999999999% (11 nines) durability.

Multiple Storage Classes

Cost-efficient options by access pattern.

Use Cases

Backup, data lakes, big data analytics, static sites.

Security Features

Access control, encryption, versioning options.

Lifecycle Policies

Automate tiering to manage long-term costs.

Data Transfer Tools

Transfer Acceleration, Snowball.

35. Amazon EBS (Elastic Block Storage) — Overview

Block Storage

Persistent volumes for EC2 instances.

Volume Types

General-purpose SSD, Provisioned IOPS, HDD tiers.

Snapshot Backup

Point-in-time copies stored in S3.

Scalability

Resize and change type without downtime.

Encryption Built-In

KMS keys secure data at rest and in transit.

Multi-Attach

io2 volumes shareable across instances.

Performance Metrics

CloudWatch tracks IOPS and throughput.

Data Persistence

Survives instance stop or termination.

Use Scenarios

Databases, CMS, enterprise apps needing block I/O.

36. Amazon VPC Fundamentals

Virtual Network

Isolated section of AWS cloud.

CIDR Control

Define custom IPv4/IPv6 address ranges.

Subnet Segmentation

Public and private subnets for security.

Routing Layer

Route tables manage traffic flow paths.

Connectivity Options

VPN, Direct Connect, VPC Peering.

Security Layers

Security groups (stateful) & NACLs.

Flow Logs

Capture network traffic for analysis and audit.

Interface Endpoints

Keeps service calls on AWS backbone.

Hybrid Ready

Extend on-prem networks seamlessly.

37. Identity and Access Management — AWS IAM Recap

Identity Control

Manage users, groups, roles centrally.

 Policy Engine

JSON policies define fine-grained permissions.

 Least Privilege

Grant only necessary access by default.

 Multi-Factor Auth

Adds extra login security layer.

 Role Delegation

Temporary credentials for services and apps.

 Access Analyzer

Detects overly permissive settings.

 Credential Reports

Encourages secure key management.

 SSO Capability

IAM Identity Center provides workforce SSO.

 Cross-Account Management

Integrates with AWS Organizations.

38. AWS Security & Compliance

Shared Model

AWS secures infrastructure; customers secure workloads.

Encryption Services

KMS, CloudHSM, S3 SSE protect data.

Threat Detection

GuardDuty, Inspector, Macie monitor anomalies.

Network Defense

WAF, Shield, Firewall Manager mitigate attack.

Compliance Ready

ISO, SOC, PCI, HIPAA attestations in place.

Security Hub

Aggregates findings across services.

Audit Report

Self-service access to compliance reports.

Best Practices

Well-Architected Security pillar guidance.

 Automated Remediation

Lambda workflows fix policy violations.

39. AWS Pricing Models

 On-Demand

Pay only for compute used, highest flexibility.

 Reserved Instances

1- or 3-year commitment for deep discounts.

 Savings Plans

\$/hr spend commitment across compatible services.

 Spot Instances

Bid on spare capacity at up to 90% savings.

 Free Tier

12-month intro + always-free usage quotas.

 Cost Tools

Cost Explorer, Budgets, Forecasts visualized.

Tagging Strategy

Allocate costs by project or department.

Marketplace Billing

Third-party charges consolidated on invoice.

Trusted Advisor Checks

Identify idle or under-used resources.

40. AWS Auto Scaling — Recap

Dynamic Scaling

Adds or removes resources automatically.

Target Tracking

Maintains metric at desired threshold.

Predictive Mode

ML forecasts future capacity needs.

Supported Resources

EC2, ECS, DynamoDB, Aurora, Lambda.

🔑 Health Remediation

Replaces failed instances automatically.

🔑 Policy Type

Simple, step, scheduled.

🔑 Cost Control

Scales down during off-peak to save money.

🔑 Integration Path

Works with ELB for traffic distribution.

🔑 Visibility

CloudWatch dashboards show scaling activity.

41. Amazon RDS (Relational Database Service) Essentials

🔑 Managed Relational DB

MySQL, PostgreSQL, Oracle, SQL Server backend.

🔑 Automated Tasks

Backup, patching, replication handled by AWS.

🔑 Multi-AZ Deploy

Synchronous standby for high availability.

🔑 Read Replicas

Offload reads, improve scaling and DR.

🔑 Storage Auto-Grow

Expands up to 128 TB without downtime.

🔑 Security Controls

Encryption, IAM auth, VPC isolation.

🔑 Aurora Option

MySQL/PostgreSQL-compatible cloud-native engine.

🔑 Use Cases

OLTP apps, SaaS backends.

42. Amazon DynamoDB

🔑 Fully Managed NoSQL

Key-value and document database.

🔑 Low Latency

Millisecond responses at any scale.

Global Tables

Multi-region, multi-master replication.

Capacity

On-demand or provisioned with auto-scale.

Streams Integration

Real-time change data capture.

Security Model

IAM fine-grained access and encryption by default.

PIT Recovery

Restore to any second in last 35 days.

Billing Basis

Pay per request or reserved throughput.

43. Amazon CloudFront — Recap

Global CDN

600+ Edge & Regional Edge locations.

Content Types

Serves static, dynamic, video, API payloads.

TLS Everywhere

Free ACM certs enable HTTPS by default.

Caching Layer

Reduces origin load and latency.

Geo Features

Restriction and targeting for regional compliance.

Edge Computing

Lambda@Edge, CloudFront Functions.

Real-Time Logs

Stream events to Kinesis or S3 for analysis.

Usage Pricing

Pay for data transfer and HTTP/HTTPS requests.

44. Amazon SQS (Simple Queue Service)

Managed Queues

Decouple microservices and serverless apps.

Queue Types

Standard (high throughput) & FIFO.

Scalability

Handles millions of messages per second.

Visibility

Prevents duplicate message processing.

Retention Period

Store messages up to 14 days.

Security Layer

SSE-KMS encrypts data at rest.

Cost Metric

Billed per API request, first million free.

Workflow Link

Triggers Lambda, ECS, or Step Functions.

45. Amazon SNS (Simple Notification Service)

Pub/Sub Service

Fan out messages to multiple protocols.

Delivery Channels

HTTP/S, Email, SMS, Lambda.

Topic Model

Publishers send once, subscribers receive many.

Message Filtering

Subscribers get only relevant payloads.

Scalability Design

Millions of messages per second capability.

Cost Basis

Pay per publish and delivery attempts.

Common Uses

Alerts, mobile push, fan-out pipelines.

46. AWS CloudFormation — Recap

Infrastructure as Code

Templates in JSON or YAML format.

Resource Coverage

200+ AWS resource types supported.

Change Sets

Preview impact before stack updates.

Stack Sets

Deploy stacks across accounts and regions.

Automation Link

Integrates with pipelines for CI/CD.

Cost Model

Free service; pay only for created resources.

Stack Policies

Protect critical resources from deletion.

47. Amazon CloudWatch

Unified Monitoring

Metrics, logs, traces in a single service.

Custom Dashboards

Real-time visualization of KPIs.

Alarms Engine

Triggers SNS, Auto Scaling actions.

Logs Insights

Query logs with SQL-like syntax.

Synthetics Canaries

Monitor endpoint availability globally.

Retention Options

Set granular log retention policies.

48. AWS CloudTrail

Audit Trail

Records every AWS API and console action.

 Storage Options

Deliver logs to S3, CloudWatch, or EventBridge.

 Organization Trails

Capture events across all accounts.

 Integrity Checks

Validate log file hash for tamper detection.

 Event Selectors

Filter by read, write, or resource type.

 Free History

90-day console view at no charge.

 Analytics Query

Use Athena to search logs ad-hoc.

 Compliance Aid

Essential evidence for audits and forensics.

49. Amazon Route 53 — Recap

 DNS Service

Translates domain names to IP addresses.

Domain Registry

Buy and manage domains within AWS.

Routing Policies

Simple, weighted, latency, failover.

Health Checks

Monitor endpoints and enable DNS failover.

Anycast Network

Global edge locations for 100% SLA uptime.

Private Zones

Internal DNS for VPC-only resolution.

Traffic Flow

Visual editor for complex routing rules.

Cost Elements

Pay per hosted zone and DNS query volume.

50. AWS Elastic Beanstalk

Platform as a Service

Deploy code; AWS handles infrastructure.

Language Support

Java, .NET, Node.js, Python, PHP, Docker.

Automated Provisioning

EC2, ALB, Auto Scaling spun up for you.

Health Dashboard

Monitors application status continuously.

Cost Principle

No extra charge; pay underlying resources only.

Ideal Users

Devs preferring minimal DevOps overhead.

Blue/Green Swap

Instant CNAME switch for new versions.

51. AWS Trusted Advisor

Best-Practice Tool

Evaluates environment across 5 pillars.

Cost Checks

Identifies idle resources and savings opportunities.

Security Checks

Flags open ports, public S3 buckets.

Performance Checks

Recommends AZ balance and instance types.

Fault Tolerance

Alerts on missing backups or multi-AZ gaps.

Service Limits

Warns before hitting API or quota ceilings.

Weekly Reports

Email summaries of new findings.

Plan Access

Full checks with Business & Enterprise Support.

52. AWS Organizations

- Multi-account management
- Consolidated billing
- Organizational units
- Service control policies
- Account automation
- Resource sharing
- Delegated admin
- Cost allocation
- Zero footprint fee

53. Elastic Container Service (ECS) — Recap

Container Orchestration

Run Docker containers at scale.

Launch Types

EC2 (managed nodes) & Fargate (serverless).

Task Definition

JSON blueprint for container settings.

Cluster Auto Scaling

Adds instances when needed automatically.

🔑 Load Balancing

ALB routes traffic by path or host header.

🔑 Use Cases

Microservices, batch jobs, APIs on containers.

54. Topics Covered Next (Notebook Roadmap)

The notebook lists these as the upcoming topics to study:

- *Amazon EKS*
- *Amazon Fargate*
- *Amazon EFS (Elastic File System)*
- *Amazon RDS Read Replicas*
- *AWS Management Console*
- *AWS Well-Architected Framework*
- *AWS Shared Responsibility Model*
- *Disaster Recovery Strategies*
- *AWS Compliance & Governance*
- *AWS Support Plans*

55. Storage Comparison: S3 vs EBS

55.1 Amazon S3

🔑 Type

Object storage.

🔑 Main Use

Files, images, videos, backups.

Simple Example

Website images, PDFs, objects/files.

55.2 Amazon EBS

Type

Block storage.

Main Use

EC2 instance hard disk.

Example

EC2 server hard drive.

55.3 EC2 + EBS Relationship

Good to remember

If EC2 is deleted, its two EBS volumes are configured to still be there and saved (depending on the "Delete on Termination" setting). EC2's own hard drive = EBS.

56. Amazon EFS & FSx

56.1 Amazon EFS (Elastic File System)

Type

File storage.

Main Use

Shared file system.

Example

Multiple EC2 servers sharing the same files/folders.

Multiple EC2 instances can access shared files like a shared folder.

56.2 Amazon FSx

(File System x — AWS Managed file system.)

Type

File storage.

Main Use

Windows / enterprise workloads.

Example

Windows File Server, shared file system.

57. S3 Glacier & AWS Backup

57.1 S3 Glacier

Glacier

Archive storage for data no longer actively used.

Type

Archive storage.

Main Use

Long-term backup.

Example

5-year-old backups.

Good to remember

Use for data you rarely access but need to keep stored for years.

57.2 AWS Backup

Type

Backup management.

 Main Use

Centralized backup.

 Example

EC2 + EBS + RDS backups.

 Good to remember

AWS Backup isn't an actual storage type itself — it's mainly a backup management service. It's a centralized place to manage backups across multiple AWS resources.

58. Snow Family & Storage Gateway — Recap

58.1 Snow Family (General)

 Type

Physical data transfer.

 Use

Huge data transfer to AWS.

 Example

100+ TB of data physically shipped to AWS.

 Good to remember

Physically transporting huge data to AWS is what enables the Snow Family use case — sending it over a regular network connection wouldn't be practical for this volume.

58.2 AWS Storage Gateway

Type

Hybrid storage.

Use

On-premises ↔ AWS.

Example

Office server backed up to AWS.

Good to remember

If a company has an on-premises server and wants to connect it with AWS storage, Storage Gateway is what's used to bridge the two.

59. Snow Family — Device Types

Snowball Edge Storage Optimized

Data transfer + storage. Best for large data migration.

Snowball Edge Compute Optimized

Data transfer + local computing. Used when AWS needs to run processing at a remote location before/while sending data back.

 Snowcone

Small and portable device for edge locations to collect and transfer data.

 Snowmobile

Extremely huge data transfer — used for legacy AWS migrations where truly massive datasets need to move.

60. Quick-Recall: Snow Family, Inspector, CloudTrail

60.1 Snow Family — One-Line Recall

- Snowball: large data
- Snowcone: small/portable data
- Snowmobile: extremely huge data

60.2 Amazon Inspector

Automatically scans AWS resources for vulnerabilities and security weaknesses.

 Ask yourself

“Vulnerability scan?” → Inspector.

60.3 AWS CloudTrail

Keeps a record/log of who used which keys or took which actions in your AWS account.

 Ask yourself

“Who did what?” → CloudTrail.

61. Quick-Recall: CloudWatch, GuardDuty, Config

61.1 Amazon CloudWatch

Monitors the performance and metrics of your AWS resources and applications.



Ask yourself

“How is it performing?” → CloudWatch.

61.2 AWS GuardDuty

Detects suspicious or malicious activity across your AWS account and workloads.



Ask yourself

“Is something malicious happening?” → GuardDuty.

61.3 AWS Config

Records and monitors the configuration of your AWS resources and checks compliance.



Ask yourself

“Is it configured correctly?” → AWS Config.

62. Quick-Recall: Trusted Advisor, SQS, SNS

62.1 AWS Trusted Advisor

Gives recommendations for optimizing your AWS account and resources.



Ask yourself

“What should I improve?” → Trusted Advisor.

62.2 Amazon SQS (Simple Queue Service)

Holds messages in a queue, letting a service or worker process them at its own pace.



Memory hook

SQS = Queue = Wait.

62.3 Amazon SNS (Simple Notification Service)

Sends one message to multiple subscribers simultaneously and fast.



Memory hook

SNS = Notification = Send.

63. Amazon EventBridge & Amazon Kinesis

63.1 Amazon EventBridge

Routes events from different AWS services/apps to the correct target or rule.



Memory hook

EventBridge = Event → Route.

63.2 Amazon Kinesis

Part of the AWS family used to collect and analyze real-time streaming data.



Memory hook

Kinesis = Real-time data stream.

64. Kinesis Data Streams, Amazon MQ & Amazon SES

64.1 Kinesis Data Streams

Collects real-time data into a stream and lets multiple applications process it.



Memory hook

Data Streams = data flows continuously.

64.2 Amazon MQ

A managed message broker service. Useful for migrating traditional messaging applications to AWS.



Memory hook

Amazon MQ = Traditional message broker.

64.3 Amazon SES (Simple Email Service)

For sending and receiving emails at scale.

65. AWS Shared Responsibility Model & AWS CAF

65.1 AWS Shared Responsibility

AWS and the customer both share responsibility for security.

AWS Responsibility ("Security OF the cloud")	Customer Responsibility ("Security IN the cloud")
- Physical data centers - Physical servers - Physical networking	- IAM users & permissions - Passwords - Security groups

Hardware / infrastructure security	- Network access rules - Customer data - Guest OS patching on EC2
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65.2 AWS CAF (Cloud Adoption Framework)

A structured framework that helps an organization successfully move and adapt from on-premises to the AWS cloud.

66. AWS CAF Perspectives, Fargate & Professional Services

66.1 AWS CAF — The 6 Perspectives

Guides a company through cloud adoption and migration to AWS across six perspectives:

Business	Business value
People	Skills & training
Governance	Policies & compliance
Platform	Technology
Security	Protection & risk
Operations	Manage & operate

66.2 AWS Fargate

A serverless compute engine for running containers, without needing to manage servers or EC2 yourself.



Memory hook

Container + no server management = Fargate.

66.3 AWS Professional Services

AWS experts and consultants who help organizations plan, build, migrate, and optimize on AWS.

67. Elastic Beanstalk, Direct Connect & CloudFormation — Recap

67.1 AWS Elastic Beanstalk

A service for easily deploying and running applications, where AWS infrastructure (load balancing, scaling) is managed for you.



Memory hook

Deploy an application without manually managing infrastructure = Elastic Beanstalk.

67.2 AWS Direct Connect

Connects your on-premises network to AWS through a dedicated private network connection.



Memory hook

On-premises → AWS dedicated link = Direct Connect.

67.3 AWS CloudFormation

Infrastructure as code — automatically creates and manages AWS resources through templates.

68. AWS Region, Well-Architected Framework & IAM Access

68.1 AWS Region

A geographical area of AWS made up of multiple isolated Availability Zones.

68.2 AWS Well-Architected Framework

A framework to design AWS workloads according to best practices — secure, reliable, efficient, and cost-effective.

68.3 AWS IAM Access (Access Analyzer)

Checks how AWS resources are being accessed or shared, especially flagging unwanted or unintended access.

69. AWS Audit Manager, AWS Artifact & Access Keys

69.1 AWS Audit Manager

Continuously collects and organizes evidence for auditing and compliance of your AWS environment.



Ask yourself

“Collect evidence for audit/compliance?” → AWS Audit Manager.

69.2 AWS Artifact

A store of AWS's own compliance documents, reports, and agreements, available to access or download.

69.3 AWS Access Key

Used to programmatically access AWS services via the CLI, API, or SDK, with authentication credentials.

70. Access Key Structure, AWS WAF & EC2 Pricing

70.1 Access Key — 2 Parts

Access Key ID

Like a username / ID.

Secret Access Key

Like a password / secret.

70.2 AWS WAF (Web Application Firewall)

Protects web applications from malicious web traffic.

Memory hook

Filter/block a web application's HTTP/HTTPS traffic = AWS WAF.

70.3 EC2 Pricing — On-Demand

- Zero use, zero pay.

- Best for short-term / unpredictable workloads.
- Relative cost: medium-high.

71. EC2 Pricing — Reserved, Spot & Dedicated

71.1 Reserved Instances

- 1- or 3-year commitment.
- Best for long-term, predictable workloads.
- Relative cost: cheapest for steady usage.

71.2 Spot Instances

- Uses AWS's spare capacity — cheap.
- Best for flexible, interruptible workloads.
- Relative cost: cheapest option overall.

71.3 Dedicated Instances

- Dedicated hardware for your instance.
- Best for compliance / isolation requirements.
- Relative cost: most expensive.

72. AWS Shield, Network ACL & Security Group

72.1 AWS Shield

Protects AWS applications/resources from DDoS attacks.



Memory hook

DDoS attack protection = AWS Shield.

72.2 Network ACL

A firewall at the subnet level of a VPC that allows or denies inbound and outbound traffic.



Memory hook

Subnet-level traffic control = Network ACL.

72.3 Security Group

A virtual firewall at the EC2 instance level that controls inbound and outbound traffic.



Memory hook

Controlling an EC2's traffic = Security Group.

73. Amazon Macie, AWS X-Ray & Amazon Redshift

73.1 Amazon Macie

Helps discover, identify, and protect sensitive data, especially PII, in Amazon S3.



Term

PII = Personally Identifiable Information.

73.2 AWS X-Ray

Traces the flow of requests inside an application to help identify performance problems and errors.



Memory hook

Finding an application request's end-to-end path / bottleneck = AWS X-Ray.

73.3 Amazon Redshift

A fully managed cloud data warehouse, mainly used to analyze large amounts of data.



Memory hook

Large data + data warehouse + analytics = Redshift.

74. Amazon Athena, RDS & DynamoDB — Recap

74.1 Amazon Athena

A serverless service that runs SQL queries directly on data stored in S3, without managing any servers.



Memory hook

S3 data + SQL query + no server management = Athena.

74.2 Amazon RDS

Relational Database Service — lets you run a traditional SQL/relational database in a fully managed way.

74.3 Amazon DynamoDB

NoSQL — no requirement for the traditional rows/columns relational structure.

75. Amazon Neptune, ElastiCache & AWS KMS

75.1 Amazon Neptune

A graph database, where the relationships between data — the connections — are the important data.

75.2 Amazon ElastiCache

An in-memory cache that keeps frequently accessed data in memory, making applications faster.

75.3 AWS KMS (Key Management Service)

A key management service for creating, managing, and controlling encryption keys.

76. AWS Cost Explorer, Budgets & Pricing Calculator

76.1 AWS Cost Explorer

Lets you visualize and analyze how much AWS spend is going where — by service, account, or usage type.



Memory hook

Cost Explorer = “Where is my money going?”

76.2 AWS Budgets

Set a spending/budget limit and receive alerts when you approach or exceed it.



Memory hook

Budgets = “How much can I spend?”

76.3 AWS Pricing Calculator

Estimate the cost of AWS resources before you actually use them.

**Memory hook**

Pricing Calculator = “How much will it cost?”

77. AWS Cost & Usage Report, Consolidated Billing & DMS

77.1 AWS Cost and Usage Report (CUR)

Provides detailed, raw information/reports on AWS usage.

**Memory hook**

CUR = Detailed billing data.

77.2 Consolidated Billing

Combines multiple AWS accounts under an AWS Organization to receive one consolidated bill.

Structure:

- Management Account
- → Account 1, Account 2, Account 3
- → One Bill

77.3 AWS DMS (Database Migration Service)

Migrates an existing database to AWS, usually while keeping the source database up and running during migration.

**Memory hook**

DMS = Database move.

78. Migration Hub, DataSync, App Migration Service & EKS

78.1 AWS Migration Hub

Tracks migration progress from multiple migration tools/services in one central place.



Memory hook

Migration Hub = Migration tracking.

78.2 AWS DataSync

Automatically and quickly transfers large amounts of data between on-premises storage and AWS storage.

78.3 AWS Application Migration Service

Migrates (lift-and-shift) servers/applications to AWS.

78.4 Amazon EKS — Recap

Elastic Kubernetes Service. Kubernetes is a system that manages containers.



Memory hook

EKS = AWS's managed Kubernetes service.

79. AWS VPN, VPC & VPS

79.1 AWS VPN (Virtual Private Network)

Used to securely connect AWS and your on-premises/company network.

VPN uses the internet, but transports traffic securely through an encrypted tunnel.

79.2 VPC (Virtual Private Cloud)

Your logically isolated virtual network within AWS.

79.3 VPS (Virtual Private Server)

A virtual server/machine. This includes a server charge, unlike VPC which is just networking.

80. VPC vs VPS & AWS Wavelength

80.1 VPC vs VPS Comparison

VPC	VPS
• Network • Virtual Private Cloud • Builds a network environment • Organizes AWS resources • Example: AWS VPC	• Server • Virtual Private Server • A virtual machine/server • Runs applications/websites • Example: EC2 is like a virtual server

80.2 AWS Wavelength

Brings AWS compute/storage resources to 5G network customers.

In Wavelength, AWS infrastructure sits at the edge of the 5G telecom network.



Main purpose

Ultra-low latency.

81. AWS WAF Flow & Cloud Fluency

81.1 AWS WAF — Recap

Web Application Firewall. Monitors and protects the incoming HTTP traffic to a website/web application.

Flow:

- User
- ↓
- AWS WAF
- ↓
- Website / Application

81.2 AWS Cloud Fluency

The ability to understand and effectively use the cloud.

This isn't an actual AWS service — it describes a level of cloud knowledge/skill.

82. AWS Outposts, Transit Gateway & CloudFormation vs CloudFront

82.1 AWS Outposts

Physically installs AWS infrastructure into your own on-premises data center.

82.2 AWS Transit Gateway

Connects multiple VPCs and on-premises networks through one central gateway.



Memory hook

Connect multiple VPCs through a central hub.

82.3 CloudFormation vs CloudFront

CloudFormation	CloudFront
• Creates/manages AWS infrastructure • Infrastructure as Code • EC2, VPC, S3, RDS • Template → infrastructure • Creates AWS resources from a template	• Delivers content fast • CDN • Images, videos, websites • CDN / Edge • Delivers a website fast to global users

83. Patch Management & Horizontal Scaling

83.1 Patch Management

Regularly updating and securing the software and operating systems on servers/computers.



Term

Patch = a software update/fix.



Systems Manager Patch Manager

Automatically patches/updates EC2 instances and manages OS patches across many servers.

83.2 Horizontal Scaling — Recap

Increasing the quantity of servers/machines.



Memory hook

Add more servers/instances → Horizontal Scaling.

84. Vertical vs Horizontal Scaling

Vertical Scaling

Increase the power of an existing server. Example: 2 CPU → 8 CPU.

Horizontal Scaling

Increase the number of servers. Example: 1 EC2 → 3 EC2.

85. Amazon Cognito & AWS STS

85.1 Amazon Cognito

A user authentication and authorization service.

Simple explanation: for a website/app's user sign-up, login, and access control management.

Memory hook

Cognito = App/website user accounts + login.

85.2 AWS STS (Security Token Service)

Provides temporary AWS credentials for a service.

All users/applications get limited-time access via STS-generated access tokens.

86. AWS Support Plans

86.1 AWS Enterprise Support

AWS's highest-level paid support plan, designed mainly for large business/mission-critical workloads.

Includes AWS's 24x7 expert technical support and proactive guidance.

86.2 All Support Plan Tiers

Basic

Free. Basic support.

Developer

Development / testing support.

Business

Production workloads + 24/7 support.

Enterprise

Mission-critical workloads + TAM (Technical Account Manager) + proactive guidance.

87. AWS Secrets Manager, Systems Manager & License Manager

87.1 AWS Secrets Manager

Securely stores and manages sensitive secrets — passwords, API keys, database credentials.

Memory hook

Store/rotate passwords or secrets = Secrets Manager.

87.2 AWS Systems Manager

Lets you administer and manage your AWS infrastructure/servers from a single service.



Memory hook

Manage/operate EC2 servers at scale = Systems Manager.



Memory hook

Patch EC2 instances = Systems Manager Patch Manager.

87.3 AWS License Manager

Manages and controls software licenses.

88. License Manager Flow & AWS OpsWorks

88.1 License Manager — Flow

Software company with software licenses → company's software → License Manager → track/control/manage licenses.

- Manage software licenses = License Manager
- Track/control software license usage = License Manager

88.2 AWS OpsWorks

Creates application configuration and management automation through Chef/Puppet.

Instead of manually configuring each server, OpsWorks can automate/manage configuration.

89. AWS CodeDeploy, Amazon QuickSight & AWS Batch

89.1 AWS CodeDeploy

A service that automatically deploys/updates application code to servers.

- Automates application deployment.
- Deploys applications to EC2 / Lambda.
- Supports blue/green deployments.

89.2 Amazon QuickSight

AWS's Business Intelligence (BI) and data visualization service.

Used to analyze data and build charts, graphs, and dashboards.

89.3 AWS Batch

A service to automatically run and manage large-scale batch jobs.

Jobs don't run immediately when queued — AWS Batch handles the process of granting/queuing jobs to run.



Memory hook

Large number of jobs + AWS Batch = batch computing job scheduler.

90. Amazon Chime & AI Language Services

90.1 Amazon Chime

Online meeting, video conferencing, and communication service.

Used by employees/teams for online meetings and calls.

90.2 Amazon Comprehend

AWS's Natural Language Processing (NLP) service.

Reads text and extracts useful information from it.

Comprehend

Read text / analyze it.

Translate

Language translation.

Transcribe

Speech → Text.

Polly

Text → Speech.

91. Amazon Rekognition, Penetration Testing & Vulnerability Scanning

91.1 Amazon Rekognition

AWS's image and video analysis service.

Uses AI to analyze images/videos and identify objects, people, faces, and activities.

Analyzing images/videos using AI

Amazon Rekognition

 Detecting faces

Amazon Rekognition

 Identifying objects in an image

Amazon Rekognition

91.2 Penetration Testing

Simulates a real attack on a system/application to find vulnerabilities and gaps.

91.3 Vulnerability Scanning

Automatically checks systems/software for known vulnerabilities.

92. AWS Config, GuardDuty & WAF — Quick Recall Trio

92.1 AWS Config

Compliance checks: is a AWS resource correctly configured or not?

 Memory hook

Configuration / compliance checks = AWS Config.

92.2 Threat Detection — GuardDuty

Continuously monitors your AWS environment and actively detects suspicious or malicious activity.

**Memory hook**

Threat detection = Amazon GuardDuty.

92.3 Web Application Filter — AWS WAF

Inspects and filters web requests, blocking harmful attacks.

**Memory hook**

Web request filtering / blocking harmful attacks = AWS WAF.

93. DDoS Protection & Application Performance Monitoring

93.1 DDoS (Distributed Denial of Service)

A distributed denial-of-service attack floods a system with traffic to overwhelm it.

**Memory hook**

DDoS protection = AWS Shield.

93.2 Application Performance Testing / Monitoring

Monitors an application's performance, errors, and latency, etc.

**AWS X-Ray**

Request tracing.

**Amazon CloudWatch**

Metrics, logs.

